

Aditya Yadav

AI/ML Engineer | +1 (631) 974-0339 | AYadav25@stevens.edu | [LinkedIn](#) | [GitHub](#)

SUMMARY

AI/ML Engineer with **3 years of experience** building scalable machine learning solutions, AI automation workflows, and **Python-based backend systems** for SaaS and enterprise products. Skilled in developing NLP pipelines, recommendation systems, predictive analytics models, and real-time inference APIs using **Python, TensorFlow, PyTorch, and FastAPI**. Experienced in deploying AI applications on AWS, optimizing model performance, and automating large-scale data workflows, improving operational efficiency, model accuracy, and system scalability.

SKILLS

- **Languages:** Python, Java, C#, JavaScript, TypeScript, Go, C/C++, SQL
- **Frontend:** React, Next.js, HTML, CSS
- **Backend & APIs:** FastAPI, Spring Boot, Node.js, Express.js, REST APIs, GraphQL, OpenAPI
- **AI/ML Frameworks:** TensorFlow, PyTorch, Scikit-learn, XGBoost, Keras
- **NLP & LLM Tools:** Hugging Face, LangChain, OpenAI APIs, FAISS, Vector Databases, RAG Pipelines
- **Data Engineering & Analytics:** Pandas, NumPy, Power BI, Matplotlib, Airflow
- **Databases:** MySQL, PostgreSQL, MongoDB, Redis, ClickHouse
- **Cloud, DevOps & MLOps:** AWS (ECS Fargate, ECR, Lambda, API Gateway, S3, CloudWatch), Docker, Kubernetes, CI/CD, GitHub Actions, MLflow, Model Monitoring
- **Tools & Workflow:** Git, Linux, Postman, pytest, Jira, Agile/Scrum

EXPERIENCE

Atlassian | AI/ML Engineer | USA | Oct 2025 – Jul 2026

- Assisted in developing AI-powered workflow automation features supporting **10K+ enterprise users** across collaboration platforms.
- Built Python preprocessing pipelines for analyzing **500K+ ticketing and operational records** on a weekly basis.
- Supported NLP workflows using **Hugging Face and LangChain**, improving semantic search relevance by **25%**.
- Tested **FastAPI endpoints handling 15K+ daily requests** for internal productivity applications.
- Contributed to vector search and retrieval workflows using **embeddings and FAISS**, reducing lookup time by **35%**.
- Automated recurring reporting and monitoring workflows using Python, reducing manual operational effort by **30%**.
- Collaborated with senior engineers to deploy and monitor **5+ AI services on AWS and Docker** environments.
- Participated in debugging, testing, and optimizing ML workflows, improving API reliability and processing performance by **20%**.

Zoho | Software Engineer | India | Apr 2022 – Aug 2024

- Developed Python automation workflows supporting CRM and analytics applications used by **100K+ customers**.
- Built scalable data processing pipelines handling **1M+ customer and operational records** across internal SaaS platforms.
- Integrated machine learning models into enterprise applications, improving workflow efficiency and task automation by **32%**.
- Enhanced **REST APIs and backend services using FastAPI and PostgreSQL** for AI-driven reporting and analytics features.
- Automated recurring reporting, validation, and operational tasks using Python scripts, reducing manual effort by **40%**.
- Optimized database queries and backend workflows using **PostgreSQL and Redis**, reducing API response times by **27%**.
- Collaborated with cross-functional teams to deliver AI-enabled automation features across multiple cloud-based business products.

EDUCATION

Stevens Institute of Technology | M.S. | Computer Science | May 2026

CERTIFICATIONS

AWS Certified Machine Learning Engineer – Associate (In Progress)

TensorFlow Developer Professional Certificate

PROJECTS

[Catalyst Map](#) — AI Event Research Platform | [GitHub](#)

- Built and deployed an AI research platform with **Next.js, TypeScript, Groq, and Tailwind CSS** to map catalyst-driven market impacts.
- Engineered a **server-side LLM pipeline** with structured schema validation, typed errors, and deterministic fallbacks.
- Added **Vitest** coverage for malformed model outputs, provider failures, and analysis data contracts.

[MorphArc](#) — Gesture-Controlled 3D Interaction | [GitHub](#)

- Built real-time hand-gesture 3D controls with **React, TypeScript, MediaPipe, Three.js, and Web Workers**.
- Engineered a **custom gesture pipeline** with hand-size normalization, hysteresis, EMA smoothing, backpressure, and damping.
- Validated camera, tracking, gesture, and rotation behavior with **66 automated tests**.

[RunRoute](#) — Intelligent Running Route Planner | [GitHub](#)

- Built and deployed a geospatial running planner with **Next.js, TypeScript, Tailwind CSS, and Mapbox GL JS**.
- Added configurable run preferences, **deterministic route candidates**, and interactive map-based start selection.
- Engineered **typed geospatial state** and responsive WebGL map lifecycle and rendering.